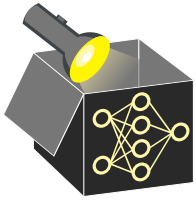
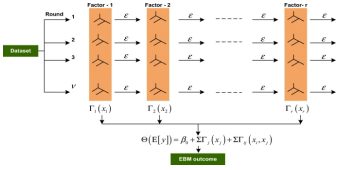
Interpretable Machine Learning



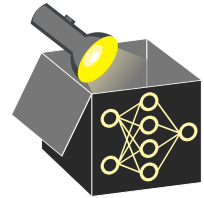
Explainable Boosting Machines



Learning goals

- Motivation from GAM
- Intelligible GAM
- Accurate GAM + Pairwise Interactions
- FAST feature interaction detection

GENERALIZED ADDITIVE MODELS

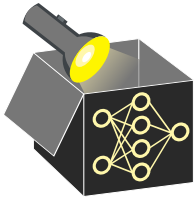


Recall idea of GAMs:

$$g(\mathbb{E}(y_i \mid \mathbf{x}_i)) = \theta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \ldots + f_p(x_{ip}), \quad i = 1, \cdots, N$$

- One shape function (spline) f_j for each feature $\rightsquigarrow$ Intelligible
- Including non-linear effects
- Better than GLMs in accuracy
- **Explainable Boosting Machines** $\leftarrow$ GAM + Gradient Boosting + Tree ensembles ▸ Lou et al. 2012
  $\rightsquigarrow$ More intelligible than unrestricted full complexity models (e.g. random forest)
  + High accuracy close to unrestricted full complexity models

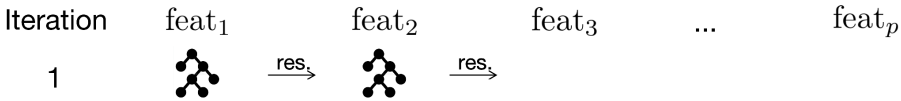
INTELLIGIBLE GAM - ALGORITHM SKETCH - PART I

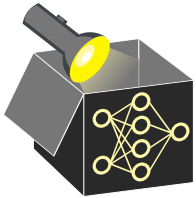


Iteration	feat_1		feat_2	feat_3	...	feat_p
1		$\xrightarrow{\text{res}_1}$				

- Set all shape functions $f_j^{(0)}$ to zero, $j = 1, \cdots, p$
- Calculate the initial pseudo-residual (i.e. the negative gradient of the loss function w.r.t. the current model prediction), e.g.
  1. for squared error loss: $\tilde{r}_i^{(0)} = -\frac{\partial L}{\partial \hat{y}_i} = y_i - \hat{y}_i^{(0)} = y_i - \sum_j f_j^{(0)}(x_{ij})$
  2. for log loss: $\tilde{r}_i^{(0)} = -\frac{\partial L}{\partial \hat{y}_i} = y_i - \sigma(\hat{y}_i^{(0)})$ with $\sigma(\cdot)$ being the sigmoid function
- Fit a shallow bagging tree with 2-4 leaves $S_1^{(1)} : x_{i1} \to \tilde{r}_i$ **only using the first feature** to predict the pseudo-residuals, with $\{x_{i1}, \tilde{r}_i\}_1^N$ as training dataset
- Update f_1: $f_1^{(1)}(x_{i1}) = f_1^{(0)} + \eta \cdot S_1^{(1)}(x_{i1})$
- Update $\hat{y}_i$: $\hat{y}_i^{(1)} = \sum_j f_j^{(1)}(x_{ij})$
- Update the residual w.r.t. the original target, e.g. $\tilde{r}_i^{(1)} = y_i - \hat{y}_i^{(1)}$lime
- Go to the second feature

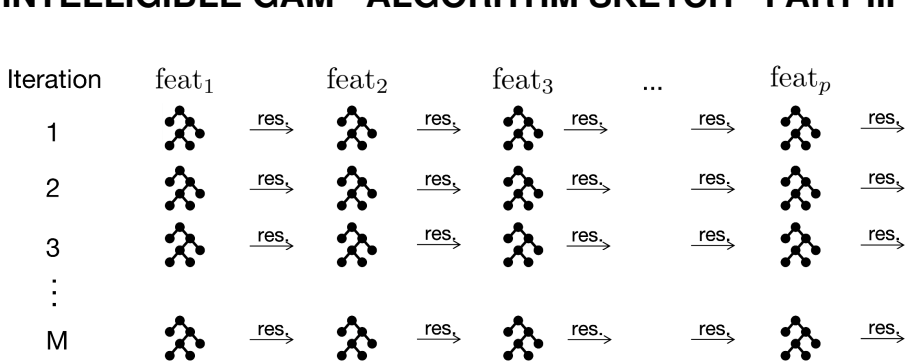
INTELLIGIBLE GAM - ALGORITHM SKETCH - PART II

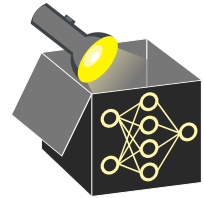




- Fit a shallow bagging tree $S_2^{(1)}$ **only using the second feature** to predict the updated residuals
- Update f_2 with $f_2^{(1)}(x_{i2}) = f_2^{(0)} + \eta \cdot S_2^{(1)}(x_{i2})$ and then update $\hat{y}_i^{(1)}$
- Update the residual: this residual takes into account both what was learned on the first tree and second tree for the first two features
- Go to feat_3

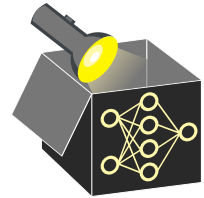
INTELLIGIBLE GAM - ALGORITHM SKETCH - PART III





- Go through every feature and keep updating the residual
- Grow a small tree on each feature but just one at a time in each iteration
- Do M iterations through these features until convergence (e.g. $M = 10000$)
  $\rightsquigarrow$ keep the learning rate η so small that feature ordering does not matter

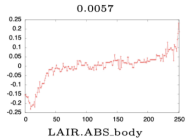
INTELLIGIBLE GAM - PREDICTION



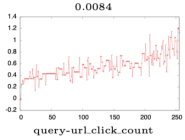
- Output: M trees which were only trained on feat_1
  + M trees which were only trained on feat_2
  $\dots$
  + M trees which were only trained on feat_p
- Summarize the weighted predictions of M trees for each unique value of feat_1
  $\hat{f}_1(x_1) = \sum_{m=1}^{M} \eta \cdot S_1^{(m)}(x_1)$, and plot $\hat{f}_1(x_1)$ vs. x_1 as a 2D graph
- Generate a 2D graph for every feature in the same way

$\rightsquigarrow$ Intelligible GAM: A series of 2D graphs as a perfect summary of predictions

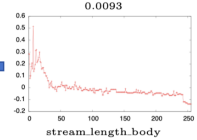
Prediction $=$



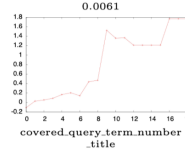
$+$



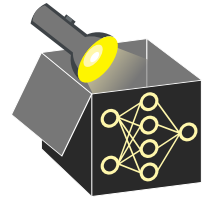
$+$



$+$ $\dots$ $+$

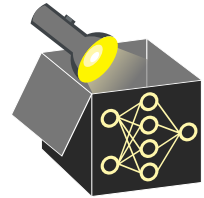


INTELLIGIBLE GAM - NOVELTY AND EXTENSION



- In each boosting step, use a bagging tree limited in size (2-4 leaves)
  - High accuracy, low variance, prevent overfitting
  - Provides discrete shape functions
  - Captures missing information in spline plots
  - High intelligibility
- Not as accurate as unrestricted full complexity model
- In the next step: ▸ Lou et al. 2013
  - Add pairwise feature interaction terms to the model
  - Replace the 2-4 regions of small tree
  - By a dynamic programming algorithm called FAST

ACCURATE GAM PLUS PAIRWISE INTERACTIONS

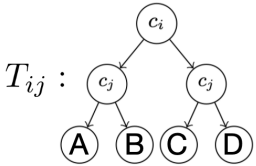


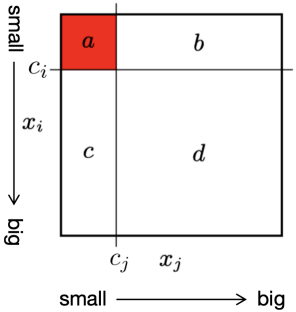
Generalized Additive Models plus Interactions:

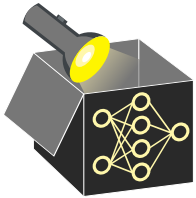
$$g(\mathbb{E}(y \mid \mathbf{x})) = \theta_0 + \sum f_i(x_i) + \sum f_{ij}(x_i, x_j) \quad \textit{for} \quad i, j = 1, \cdots, p, \quad i \neq j$$

- $O(p^2)$: a large number of pairwise interactions if p is big
  $\rightsquigarrow$ Select only a small number of the most important interactions
- Intuitively, significant reduction in RSS $\rightarrow$ Strong pair interaction
- **FAST** algorithm does quick ranking without fully building every interaction function

FAST FEATURE INTERACTION DETECTION - PART I





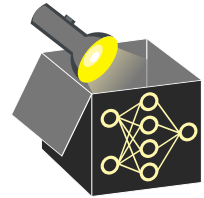


- T_{ij}: A tree-like interaction predictor with 4 leaves for each pair of features
- c_i, c_j: One cut (grid point) on the **sorted** values of feature x_i and x_j, respectively
- A, B, C, D: Notation for 4 leaf regions
- **Idea**: Search for all possible (c_i, c_j) and pick the best T_{ij} with the lowest RSS
- Prediction value on region r, $r \in \{A, B, C, D\}$:

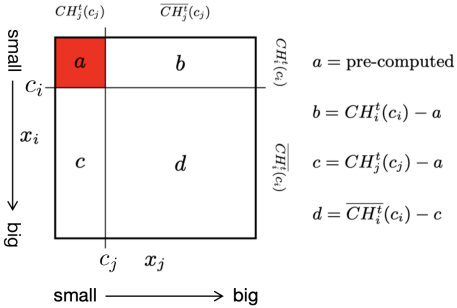
$$\hat{y}_r = T_{ij}.r = \frac{1}{|r|} \sum_{(\mathbf{x},y)\in r} y = \frac{\text{Sum of targets in } r}{\text{Sum of counts in } r}$$

- $$\begin{aligned}\text{RSS} &= \textstyle\sum_{k=1}^{N} (y_k - T_{ij}(\mathbf{x}_k))^2 \\ &= \textstyle\sum_{k=1}^{N} y_k^2 - 2\sum_r \left(\hat{y}_r \cdot \sum_{(\mathbf{x},y)\in r} y\right) \\ &\quad + \textstyle\sum_r \hat{y}_r^2 \cdot |r|\end{aligned}$$

FAST FEATURE INTERACTION DETECTION - PART II



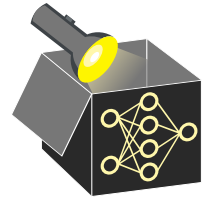
- How to quickly get sum of targets and sum of counts for all possible cuts?
  $\rightsquigarrow$ Use **marginal cumulative histograms** (CH)



An illustration for computing sum of targets for each quadrant

- For example:
  $CH_j^t(c_j)$: cumulative sum of targets for data points with $x_j \leq c_j$;
  $\overline{CH_j^t}(c_j)$: cumulative sum of targets for data points with $x_j > c_j$
- Create **two lookup tables** to store values for every (c_i, c_j), :
  1. Compute the cumulative sum of targets in four regions, and store the values in the **table of targets** $L^t(c_i, c_j) = [a, b, c, d]$
  2. Compute the cumulative sum of counts in four regions, and store the values in the **table of counts** $L^w(c_i, c_j) = [a, b, c, d]$
- Given any (c_i, c_j), values of (b, c, d) can be easily derived with known a
  $\rightsquigarrow$Construct lookup tables using dynamic programming

FAST FEATURE INTERACTION DETECTION - PART III



- For x_i with d_i possible values and x_j with d_j possible values:
  $\rightsquigarrow$ compute and store $d_i d_j$ records of $[a, b, c, d]$ in both table of targets and table of counts
- For each pair of (x_i, x_j), calculate RSS for each possible (c_i, c_j) and pick the best T_{ij} with the lowest RSS [Note: $\hat{y}_A$ can be simply calculated as $L^t(c_i, c_j).a / L^w(c_i, c_j).a$]

$d_i d_j$ possible (c_i, c_j)

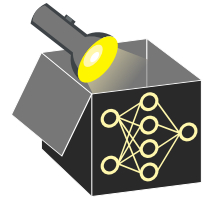
a	b	$\Rightarrow$	$\hat{y}_A$	$\hat{y}_B$	$\Rightarrow$	RSS_1
c	d		$\hat{y}_C$	$\hat{y}_D$		
a	b	$\Rightarrow$	$\hat{y}_A$	$\hat{y}_B$	$\Rightarrow$	RSS_2
c	d		$\hat{y}_C$	$\hat{y}_D$		
$\vdots$			$\vdots$			$\vdots$
a	b	$\Rightarrow$	$\hat{y}_A$	$\hat{y}_B$	$\Rightarrow$	$RSS_{d_i d_j}$
c	d		$\hat{y}_C$	$\hat{y}_D$		

select the lowest RSS and assign it as the weight for (x_i, x_j) to measure the strength of interaction

FAST FEATURE INTERACTION DETECTION - PART IV

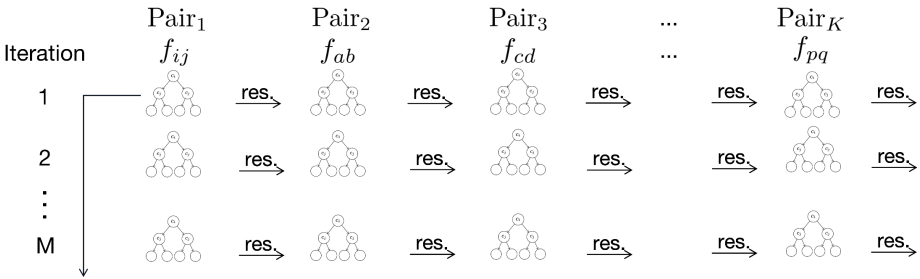
- Complexity analysis:
  Computing cumulative histograms: $O(N)$ (N: number of observations)
  Constructing matrices of sum values: $O(N + d_i d_j)$
  $\rightsquigarrow$ Complexity per pair: $O(N + d_i d_j)$
- Discretize continuous feature values in b bins ($b \leq 256$)
  $\rightsquigarrow$ $O(N + b^2)$ per pair
  $\rightsquigarrow$ Choose optimal number of bins b w.r.t. bias-variance trade-off
- Quick ranking and select top K pairs

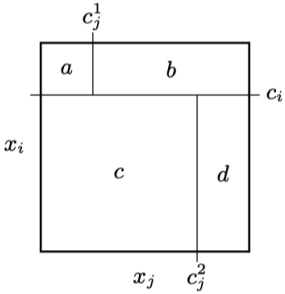
ACCURATE GAM - TWO-STAGE CONSTRUCTION

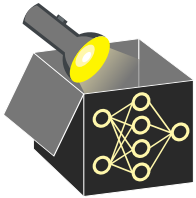


1. Fit main effects
   - Fit model only using univariate terms
   - Fix the main effects model
2. Fit pairwise residual of main effects model to original targets
   - Use FAST to sort through $O(p^2)$ pairs
   - User selects number K of pairs to be added
   - Run boosting algorithm to fit K pairs
   - Final model = p mains + K pairs

ACCURATE GAM - 2. STAGE ALGORITHM SKETCH

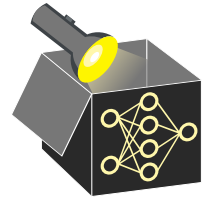




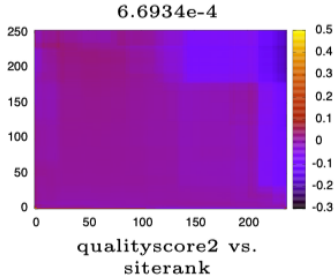


- To model the pairwise interactions effectively and efficiently ⇝Replace the shallow tree ensembles used in Intelligible GAM with a tree-like predictor (similar to FAST)
- 3 cuts instead of 2 cuts
- Re-use the stored values in the lookup tables for sum of targets and sum of counts from FAST
- Greedy search for the best combination of cuts (lowest RSS)

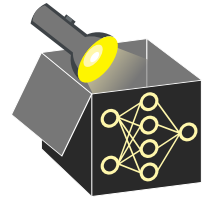
ACCURATE GAM - PREDICTION



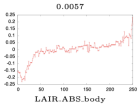
- Output: M predictors which were only trained on Pair_1
  + M predictors which were only trained on Pair_2
  . . .
  + M predictors which were only trained on Pair_K
- Summarize the weighted predictions of M models in a 3D heatmap
- The dimension of color represents the contribution of pairwise feature values to the prediction
- Generate a 3D heatmap for each pair of features



EBM - FINAL MODEL

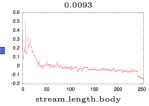


Main_1 feat_1 | Main_2 feat_2 | Main_3 feat_3 | ... ... | Main_p feat_p

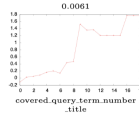


+

+

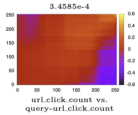


+ ... +

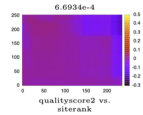


Prediction =

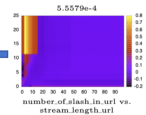
+



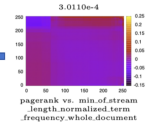
+



+



+ ... +



Pair_1 f_{ij} | Pair_2 f_{ab} | Pair_3 f_{cd} | ... ... | Pair_K f_{pq}